Materials and Methods

Ancestry sorting and genomic architecture

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Module A (_PK pipeline)

Working draft for the _PK pipeline. It covers the extent, direction and genomic architecture of ancestry sorting, using the LD-reduced units produced in Module 0 (LD decay and complexity reduction). Climate association and dependence-aware (block-bootstrap) uncertainty are documented separately. Blue boxes mark items that remain open.

S1 Scope and analytical principles

These methods describe the quantification of the extent and direction of ancestry sorting and the analysis of its genomic architecture, using the LD-reduced units from Module 0 (LD decay and complexity reduction).

Three principles guided the analyses. First, markers in LD carry partly redundant information. We therefore used clusters of strongly correlated markers as LD-reduced analytical units wherever possible, while retaining marker-level analyses as diagnostics of spatial pseudoreplication. We do not assume that the resulting units are strictly statistically independent. Second, a variable evaluated as a predictor was not also used to select the analysed data. In particular, loci were gated by parental allele frequency rather than by the diagnostic index (DI), leaving the full range of DI available for analysis. Third, these analyses do not assume complete demographic independence of the twenty hybrid populations; analyses that use variation among populations are interpreted as descriptive associations.

Open item. The general sample description currently reports 165 sequenced hybrid individuals, whereas the analysis objects contain 164. Confirm which individual was excluded, most likely because of excessive missing data, and report the identifier and exclusion criterion in the final text. Until then, 164 is used for all analysis-specific counts.

S2 Extent and direction of ancestry sorting

S2.1 Near-fixation and ancestry orientation

For each marker or cluster consensus, dosage was oriented as the *F. aquilonia*-allele frequency using the sign of the parental allele-frequency difference. Units for which the parents did not differ had no defined parental direction. Within each hybrid population, a unit was classified as near-fixed toward *F. aquilonia* when the oriented allele frequency was at least ϕ, near-fixed toward *F. polyctena* when it was at most $1 - \phi$, and otherwise not near-fixed. Here $\phi = 0.85$ is the frequency the fixed-for (major) allele must reach, equivalently the requirement that the opposing allele fall to at most $1 - \phi = 0.15$. The same threshold was used throughout because the attainable frequency values depend on the number of sampled diploid individuals in a population.

S2.2 Sorting magnitude and direction

Let n_{obs} be the number of populations with an observed and oriented frequency, and n_{aqu} and n_{pol} the numbers near-fixed toward *F. aquilonia* and *F. polyctena*, respectively. Sorting magnitude was the total proportion near-fixed,

$$p_{\text{fixed}} = \frac{n_{\text{aqu}} + n_{\text{pol}}}{n_{\text{obs}}}. \tag{1}$$

A unit passed the sorting-magnitude gate when $p_{\text{fixed}} \geq \tau = 0.6$. Among units passing this gate, directional consistency was assessed by an exact two-sided binomial test of n_{aqu} successes in $n_{\text{fixed}} = n_{\text{aqu}} + n_{\text{pol}}$ trials under a random-direction null probability of 0.5. A significant imbalance ($P < 0.05$) defined unidirectional sorting, with the parental direction set by the larger count. A non-significant split was classified as bidirectional when n_{fixed} was sufficient for the test to reject an all-one-parent outcome under the null; otherwise it was labelled ambiguous. Units failing the magnitude gate were unsorted. Exchanging parental labels changes the direction but not sorting magnitude or class.

The two thresholds were treated separately. The near-fixation floor ϕ is the per-population convention; the sorting threshold τ is the cross-population comparison. Their influence on the results was evaluated over a grid of $\tau \in \{0.5, 0.6, 0.7, 0.8\}$ and $\phi \in \{0.80, 0.85, 0.90, 0.95\}$: the estimated prevalence of sorting declines monotonically with either threshold, whereas the diagnostic-index direction reversal is preserved (threshold-sensitivity supplement, Figs S1–S2).

Nouhaud et al. [1] evaluated predictable ancestry sorting by asking whether the same ancestry component repeatedly fixed across three hybrid populations and summarising the outcome in fixed 20-kb windows. The present classification retains that biological contrast but extends it to twenty populations, estimates the outcome directly from population allele frequencies, and separates the prevalence of near-fixation from evidence that its parental direction is repeated more consistently than expected under random direction. Applying the statistic first to SNPs and then to LD-reduced units also avoids assigning equal analytical status to physical windows that may contain very different amounts of redundant information.

S2.3 Parental polymorphism gate

Near-fixation was interpreted as sorting only for variants segregating in the founding parental pool. Analyses were therefore restricted to units with pooled parental MAF ≥ 0.15, calculated from pooled parental allele counts and folded to the minor allele. This gate excludes variants already close to monomorphic in the parental pool without selecting on DI, which is evaluated as a predictor in subsequent analyses.

S2.4 Cluster-level analysis and dilution check

The sorting statistic was first calculated per marker. It was then recalculated for eMLG consensus genotypes so that a cluster contributed one aggregate observation. For clusters without a stored consensus but containing a marker-level sorting signal, a consensus was constructed by the same procedure used for larger clusters. Parental consensus genotypes reused the reference marker and allele-flip decisions derived from the hybrid genotypes, preventing hybrid and parental consensuses from being placed on inconsistent orientations. Cluster DI was aggregated as the maximum DI among member markers.

The cluster-level aggregation analysis is conditional: clusters entered this comparison because they contained at least one marker with a SNP-level sorting signal. Consequently, the proportion of aggregated clusters classified as unsorted does not estimate the genome-wide fraction of unsorted clusters. Instead, it asks whether an apparent marker-level sorting signal remains after correlated members are represented jointly.

When a cluster containing an individually sorted marker was unsorted after aggregation, consensus fidelity was used to distinguish lack of an aggregate signal from possible dilution in a heterogeneous consensus. Low-fidelity clusters were flagged rather than automatically interpreted as evidence against sorting.

S2.5 Models of sorting direction

Among unidirectionally sorted units, the probability of sorting toward *F. aquilonia* was modelled by logistic regression. The genome-wide direction model is described with the architecture analyses below. The robustness of the diagnostic-index direction reversal to the classification thresholds is shown in the threshold-sensitivity supplement (Figs S2 and S3). Associations are interpreted descriptively and are not assigned a causal mechanism from these regressions alone.

S3 Genomic architecture of sorting

S3.1 Marker-level genomic summaries

Map-based recombination rates were assigned to markers by linear interpolation and expressed in cM/Mb. For each retained marker, let p_{aqu} and p_{pol} denote the allele frequencies in the *F. aquilonia* and *F. polyctena* reference samples, respectively, and let

$$\bar{p} = \frac{p_{\text{aqu}} + p_{\text{pol}}}{2}.$$

We calculated the mean expected heterozygosity within the parental species as

$$H_S = \frac{2p_{\text{aqu}}(1 - p_{\text{aqu}}) + 2p_{\text{pol}}(1 - p_{\text{pol}})}{2},$$

and expected heterozygosity based on the equally weighted pooled parental allele frequency as

$$H_T = 2\bar{p}(1 - \bar{p}).$$

The between-parent allelic-difference summary was calculated as

$$d_{xy} = p_{\text{aqu}}(1 - p_{\text{pol}}) + p_{\text{pol}}(1 - p_{\text{aqu}}),$$

and relative parental differentiation as

$$F_{\text{ST}} = \frac{H_T - H_S}{H_T},$$

for markers with $H_T > 0$.

These quantities are marker-level summaries conditional on the polymorphic sites retained for analysis. Because invariant sites were not included, H_S and d_{xy} should not be interpreted as genome-wide nucleotide diversity or absolute sequence divergence, respectively.

S3.2 Marker-level and LD-reduced summaries

Sorting was summarised across recombination-rate deciles twice: once using one representative per LD-reduced unit and once using a random sample of individual markers. This comparison was used to diagnose spatial pseudoreplication. Differences between the two summaries indicate that a marker-level pattern is influenced by unequal redundancy across the recombination landscape.

S3.3 Association models

Relationships among DI, recombination rate, parental expected heterozygosity, between-parent allelic difference, relative differentiation, pooled parental MAF, and cluster size were first summarised using rank correlations and recombination deciles (architecture results summary, Fig. 1a).

The magnitude of sorting was modelled against recombination rate at the LD-reduced-unit level. Cluster size was not included in this model because it can be a consequence of both recombination and local marker variability; conditioning on it could induce an association between its causes.

Among unidirectionally sorted units, sorting direction was analysed by logistic regression with standardised DI, standardised $\log_{10}$(recombination + 0.1), standardised pooled parental MAF, and standardised log cluster size as predictors (architecture results summary, Fig. 1c). The recombination coefficient is treated as an association rather than as evidence of linked selection. Coefficient sensitivity will be regenerated with the same binomial direction rule before it is included in the supplementary results.

S3.4 Consensus-versus-representative validation

For clusters represented by both an eMLG consensus and a representative marker, we compared the inferred sorting direction between representations. The two representations agreed on the parental direction for approximately 99% of these clusters, confirming that using representatives for genome-wide architecture models does not materially change the parental direction assigned to clusters, while the consensus remains preferable for aggregate sorting magnitude and cluster-level LD analyses.

S4 Statistical interpretation

The analyses are descriptive and conditional on the sampled populations, quality gates, and analysis definitions. Consistent sorting can be produced by more than one evolutionary process; distinguishing drift from selection requires the recombination-matched neutral null, because a non-recombining autosomal block does not have lower N_e than its surroundings under neutrality. The weak residual low-recombination pull toward *F. aquilonia* that survives control for DI is therefore reported as a lead for the intrinsic hypothesis rather than as evidence of linked selection.

S5 Software, reproducibility, and current status

Ancestry-sorting and architecture analyses were performed in R using scripts in the `formica_hybrid` repository, folder `moduleA_sorting/R/`: `moduleA_sorting_phenomenon.R` (extent and direction of sorting, eMLG consensus construction, dilution check, and the τ/ϕ threshold-sensitivity sweeps) and `moduleA_architecture.R` (genomic architecture and sorting-direction models); shared statistical functions are in `parallelism_stats.R` and `eMLG_parallelism.R`. Ohta-statistic functions used elsewhere in the project are supplied by `LDscnR`. Shared genotype inputs and the Module 0 clustering are read from `data/` and `module0_ld_pruning/data/`; module intermediates are cached to `moduleA_sorting/data/` with a parameter-stamped validity check so that reruns reuse them when settings are unchanged; figures are written to `moduleA_sorting/Figures/`.

Open item. Add the final `LDscnR` citation, version, and repository or archive identifier before submission.

Table 1: Principal parameter settings used in the Module A analyses.

Analysis	Parameter	Value
Sorting	Near-fixation floor	$\phi = 0.85$ (fixed-for allele)
	Sorting-magnitude threshold	$\tau = 0.6$
	Direction test	exact binomial, null probability 0.5, $\alpha = 0.05$
	Parental MAF gate	≥ 0.15
	Cluster DI aggregation	maximum over members
Threshold sensitivity	Sorting threshold grid	$\tau \in \{0.5, 0.6, 0.7, 0.8\}$
	Near-fixation floor grid	$\phi \in \{0.80, 0.85, 0.90, 0.95\}$
Architecture	DI treatment	continuous, ungated covariate
	Direction model	logistic: DI, $\log_{10}$(recomb+0.1), parental MAF, log size
	Magnitude model	prop_fixed ∼ recombination (+ DI)

References

[1] Nouhaud P, Martin SH, Portinha B, Sousa VC, Kulmuni J. Rapid and predictable genome evolution across three hybrid ant populations. *PLoS Biology.* 2022;20:e3001914. doi:10.1371/journal.pbio.3001914.